Aspirin (Acetylsalicylic Acid): Structure, Synthesis, Reactivity, and Pharmacology

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CHEM 332 / Organic Chemistry
Final Project — Individual Submission

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1: Compound Chosen

Acetylsalicylic acid (**aspirin**), molecular formula $C_9H_8O_4$, molar mass 180.16 $g\,mol^{-1}$. It is the acetyl ester of salicylic acid and is one of the most widely used small-molecule drugs in the world, taken both for analgesia and as a low-dose antiplatelet agent.

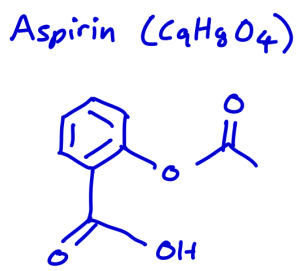

Figure 1: CHEM 332_Scheme 1. Skeletal structure of aspirin, $C_9H_8O_4$.

2A: Structural Analysis

Resonance Structures

The chemically important set of contributors comes from the acetoxy group's ester oxygen donating its lone pair into the aromatic ring (Figure 2). The lone pair on O can either donate into the ester C=O (R−Ö−C=O ↔ R−O$^+$=C−O$^-$) or push *through* the aryl C–O bond into the ring's π system, placing negative charge at the ortho and para carbons (C3 and C5). Two consequences: the acetoxy group acts as a weak π-donor / o,p-director in EAS even though it is inductively electron-withdrawing (Section C), and the ester carbonyl is slightly less electrophilic than a comparable ketone. The benzene Kekulé forms and the standard C=O ↔ C$^+$−O$^-$ carbonyl resonance of the carboxylic acid contribute as well.

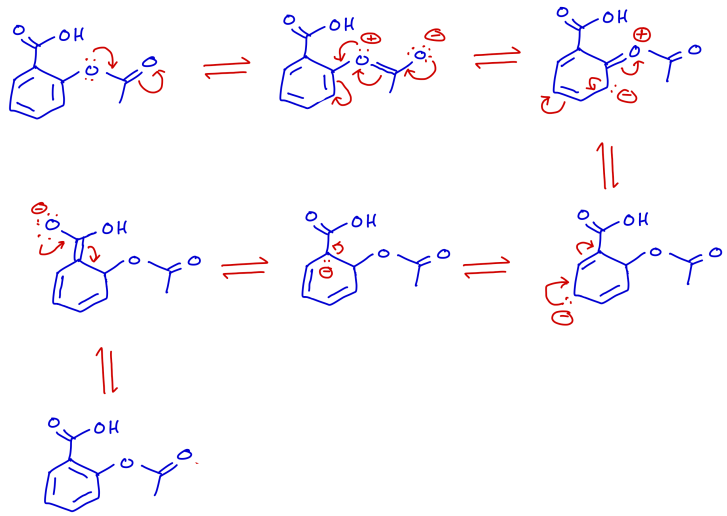

Figure 2: CHEM 332_Scheme 2. Resonance contributors of aspirin showing acetoxy-oxygen lone-pair donation into the aromatic ring. The π-donor character of the OAc group is what activates the C3 and C5 positions and makes it an o/p director in EAS, despite its inductively electron-withdrawing σ effect.

Functional Groups

Aspirin is a benzene ring bearing two ortho substituents: a **carboxylic acid** (−COOH) and an **aryl acetate** ester (−OC(O)CH_3). Counting the aromatic ring itself, the molecule has three distinct functional groups, each of which controls a different piece of its reactivity (Figure 3).

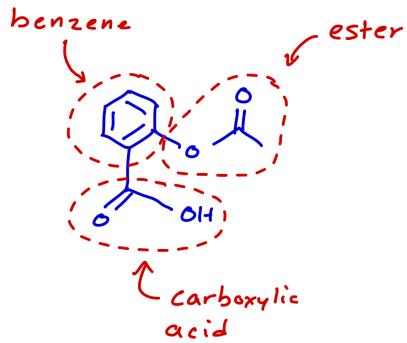

Figure 3: CHEM 332_Scheme 3. Functional groups in aspirin: the aromatic ring (benzene), the aryl acetate ester at C2, and the carboxylic acid at C1.

Chiral Centers

Aspirin has **no chiral centers**. No carbon in the molecule carries four different substituents, so there are no stereocenters and no R/S configurations to assign; the molecule is achiral.

B. Synthesis Pathway

The standard laboratory synthesis is the Fischer-type acylation of salicylic acid with acetic anhydride under acid catalysis, with acetic acid as the byproduct (Scheme 4). Conditions are typically 50–60 °C with a catalytic amount of concentrated H_2SO_4 (or H_3PO_4 in the classroom version) and a slight excess of acetic anhydride to push the equilibrium toward product. After the reaction, the mixture is quenched with cold water, which destroys the excess anhydride and precipitates the much less-soluble aspirin for vacuum filtration and recrystallization from ethanol/water.

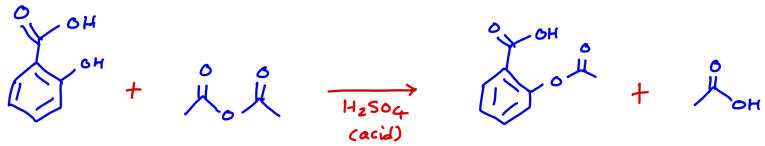

Figure 4: CHEM 332_Scheme 4. Overall reaction: salicylic acid + acetic anhydride $\xrightarrow{H_2SO_4}$ aspirin + acetic acid.

The reaction proceeds in three mechanistic stages. To save space the acid catalyst is abbreviated HA.

Activation of Acetic Anhydride

The acid protonates one of the carbonyl oxygens of acetic anhydride. This raises the LUMO of the carbonyl carbon and turns the anhydride into a much better electrophile (Figure 5). Without this activation the phenolic −OH of salicylic acid is too weak a nucleophile to react at a useful rate.

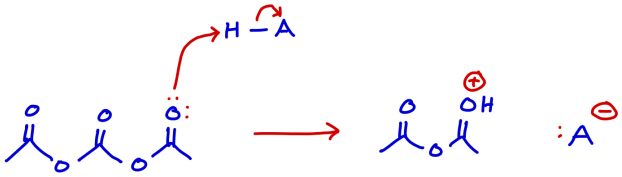

Figure 5: CHEM 332_Scheme 5a. Acid activation of acetic anhydride.

Nucleophilic Attack by Salicylic Acid

The phenolic oxygen of salicylic acid attacks the activated carbonyl carbon, producing a tetrahedral intermediate. The conjugate base of the catalyst removes the resulting oxocarbenium proton (Figure 6). Note that the phenolic −OH reacts preferentially over the carboxylic −OH: the carboxylic oxygen lone pairs are tied up in resonance with the adjacent C=O and so are less nucleophilic.

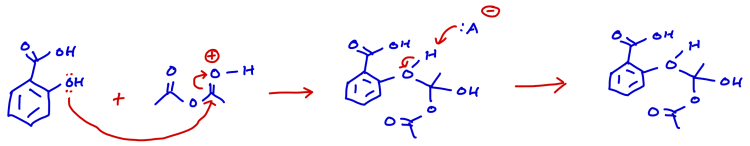

Figure 6: CHEM 332_Scheme 5b. Nucleophilic addition of salicylic acid onto the activated anhydride to give the tetrahedral intermediate.

Acetate as a Leaving Group and Deprotonation

A proton transfer onto the acetate leaving group converts it into neutral acetic acid; collapse of the tetrahedral intermediate then expels acetic acid and regenerates the carbonyl. A final deprotonation yields aspirin and returns the catalyst (Figure 7). The acid is therefore strictly catalytic: it is consumed in step 1 and regenerated by the end of step 3.

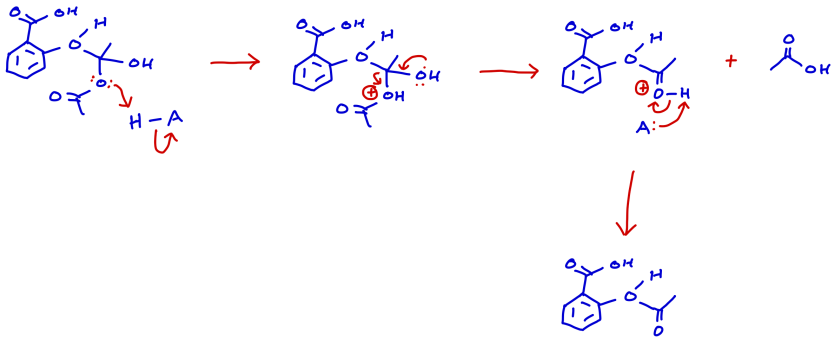

Figure 7: CHEM 332_Scheme 5c. Proton transfer, collapse of the tetrahedral intermediate, and expulsion of acetic acid to deliver aspirin.

C. Functional Group Reactivity

Each of aspirin's three functional groups has one characteristic reaction from Organic I/II. The base-catalyzed hydrolysis of the ester (below) is also the reaction I work through in detail with a full mechanism, because it is mechanistically identical to the COX-1 acetylation step in Section D.

Ester – Hydrolysis

The acetyl ester undergoes both acid- and base-catalyzed hydrolysis back to salicylic acid and acetic acid; this is what limits aspirin's shelf life in humid storage and in aqueous formulations. Mechanistically (Figure 8): (1) HO^- attacks the electrophilic ester carbonyl, breaking the C=O π bond and forming a tetrahedral alkoxide; (2) the alkoxide collapses, reforming the carbonyl and expelling salicylate as the leaving group (a competent leaving group because the resulting negative charge is stabilized both inductively by the adjacent ring carboxyl and by delocalization into the ring); (3) a fast acid–base equilibration delivers neutral salicylic acid and acetic acid. Under basic conditions the reaction is overall irreversible because acetate is deprotonated to its conjugate base, so acid-catalyzed re-esterification cannot restart the cycle.

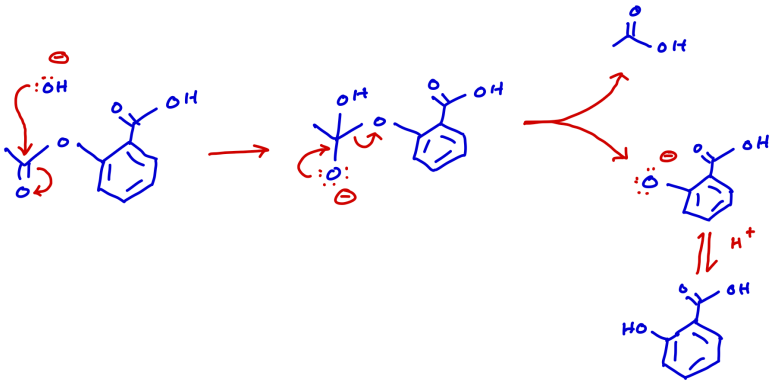

Figure 8: CHEM 332_Scheme 6. Base-catalyzed hydrolysis of aspirin's acetyl ester: (1) HO^- attacks the ester carbonyl, giving a tetrahedral alkoxide; (2) collapse expels salicylate and gives acetate; (3) proton transfer delivers neutral salicylic acid and acetic acid.

Benzene – Electrophilic Aromatic Substitution

The aromatic ring is overall deactivated relative to benzene (both substituents are net electron-withdrawing inductively) but is still capable of EAS. The two substituents direct cooperatively (Figure 9): the carboxylic acid is an EWG / meta-director, and the acetoxy is a weak π-donor / o,p-director through its oxygen lone pair (the resonance shown in Figure 2). Both substituents therefore activate the same set of positions, and the **C5 position** (para to −OAc, meta to −COOH) is the kinetically preferred site because it sits furthest from the steric bulk of the two ortho substituents.

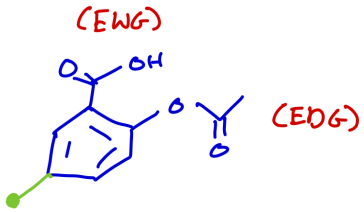

Figure 9: CHEM 332_Scheme 7. Directing analysis for EAS on aspirin. −COOH (EWG, meta-director, top) and −OAc (weak EDG, o/p-director, right) both activate the C3 and C5 positions; C5 (highlighted in green) is the kinetically preferred site.

Carboxylic Acid – Acid/Base Reactivity & Deprotonation

The −COOH group (p$K_a \approx 3.5$) is substantially deprotonated at physiological pH (7.4) to give the carboxylate (Figure 10). The carboxylate is what dissolves in plasma and what makes the electrostatic contact with Arg120 in the COX active-site mouth (Section D).

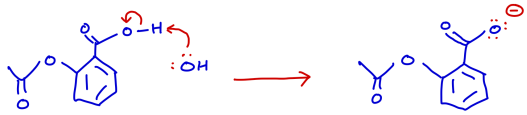

Figure 10: CHEM 332_Scheme 8. Acid–base reactivity of aspirin's carboxylic acid: hydroxide deprotonates the −COOH to give aspirin carboxylate (acetylsalicylate) and water.

D. Biological / Pharmacological Role

Aspirin's pharmacology is mechanistically tied to one specific reactive feature of its structure: the **acetyl ester is a covalent acetylating agent**. Aspirin diffuses into the active-site channel of cyclooxygenase (COX-1 and COX-2), where Ser530 is positioned next to the acetyl carbonyl. Ser530 nucleophilically attacks the carbonyl carbon (essentially the same nucleophilic acyl substitution as the hydrolysis step in Section C), aspirin's acetyl group is transferred onto the serine, and salicylate is released as the leaving group. The serine is now permanently acetylated and cannot bind arachidonic acid, so prostaglandin and thromboxane synthesis is blocked.

This explains aspirin's two clinical uses:

- **Anti-inflammatory / antipyretic / analgesic.** Blocking prostaglandin H_2 synthesis lowers downstream PGE_2, which sensitizes peripheral nociceptors and drives the hypothalamic fever response. Less PGE_2 means less pain, less swelling, and a lower temperature set point.
- **Antiplatelet / cardioprotective.** Platelets cannot synthesize new protein, so once their COX-1 is acetylated they stay inhibited for their lifetime (∼7–10 days). With thromboxane A_2 suppressed for that long, platelet aggregation is blunted – the basis of the daily 81 mg "baby aspirin" regimen for secondary prevention of myocardial infarction and ischemic stroke.

The structural features driving this are exactly the ones identified in Section A: the ester provides the transferable acetyl group, the carboxylate ($-COO^-$ at physiological pH) anchors aspirin in the polar mouth of the COX channel via Arg120, and the aromatic ring orients the molecule so Ser530 sits in line with the acetyl carbonyl.

E. Safety and Environmental Impact

Safety Concerns

At therapeutic doses (325–650 mg for analgesia, 75–100 mg daily for cardioprotection) aspirin is well tolerated. The most common adverse effect is gastrointestinal: by inhibiting COX-1 in the stomach lining it reduces the protective prostaglandin layer and raises the risk of gastric irritation, ulcers, and upper-GI bleeding with chronic use. Salicylate overdose (>150 mg/kg) produces tinnitus, mixed respiratory-alkalosis/metabolic-acidosis disturbance, hyperthermia, and

in severe cases coma; treatment is urinary alkalinization with sodium bicarbonate and, if severe, hemodialysis. Aspirin is also contraindicated in children with viral illness because of the association with Reye's syndrome.

Environmental Impact

Aspirin and its primary metabolite salicylic acid enter wastewater through human excretion and improper disposal. They are biodegradable and do not strongly bioaccumulate, but chronic low-level exposure has been documented to cause oxidative stress and histopathological changes in the livers of fish (e.g. *Labeo rohita*) at environmentally relevant concentrations. Aspirin is on the WHO Model List of Essential Medicines, and manufacturing waste is regulated under standard pharmaceutical-effluent rules.

F. Real-World Application

In a February 2026 *RSC Mechanochemistry* paper, Garumanna, Thakuria, and Adassooriya prepared aspirin **nanocrystals** ($\sim$150 nm) by liquid-assisted ball-milling of bulk aspirin with water at 30 Hz for 30 min, which is essentially a top-down, solvent-minimal modification of the molecule's solid form rather than its covalent structure. The nanocrystals released $\sim$90 % of the drug within 6.5 min at simulated gastric pH ($\sim$1.2), versus $\sim$50 % for bulk aspirin under the same conditions, and the authors track a polymorph transition (Form IV $\rightarrow$ Form I) during milling. Practically, this matters because aspirin's oral bioavailability is normally only $\sim$50 % on account of its poor aqueous solubility – a $\sim 3\times$ acceleration of dissolution at gastric pH is exactly the kind of formulation gain that translates into faster onset of action without changing the chemistry of the molecule itself.

References:

[1] Garumanna, G. D. S. K.; Thakuria, R.; Adassooriya, N. M. Mechanochemical synthesis of aspirin nanocrystals for pharmaceutical applications. *RSC Mechanochem.* **2026**, Advance Article. `https://doi.org/10.1039/D5MR00118H`.

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[3] Awtry, E. H.; Loscalzo, J. Aspirin. *Circulation* **2000**, *101* (10), 1206–1218. `https://doi.org/10.1161/01.CIR.101.10.1206`.

[4] Gayen, T.; Tripathi, A.; Kumari, U.; Mittal, S.; Mittal, A. K. Ecotoxicological impacts of environmentally relevant concentrations of aspirin in the liver of *Labeo rohita*: biochemical and histopathological investigation. *Chemosphere* **2023**, *333*, 138921. `https://doi.org/10.1016/j.chemosphere.2023.138921`.

[5] Clayden, J.; Greeves, N.; Warren, S. *Organic Chemistry*, 2nd ed.; Oxford University Press: Oxford, 2012; Chapters 10 (carbonyl chemistry), 22 (electrophilic aromatic substitution).

Reflections

1. Approach. I worked through the rubric in order so that each section could feed into the next: structure first, then synthesis (to put the resonance and functional-group analysis to use), then reactivity (which set up Section D's enzymology), and finally safety and recent literature. Clayden, Greeves, and Warren was my main reference for the carbonyl and EAS chapters, while Vane and Botting's COX-inhibition review tied the pharmacology directly back to the acetylation chemistry I had already drawn out. Drawing every figure in ChemDraw before writing the prose forced me to commit to a mechanism early, which made the prose much faster to write afterwards.

2. Challenge. The hardest part was justifying the EAS regiochemistry, since aspirin's two ring substituents are *ortho* to each other and push in different directions ($-$COOH is an EWG / meta director while $-$OAc is a weak π-donor / o,p director). I drew the σ-complex resonance forms for electrophile addition at every ring position by hand and confirmed that only C3 and C5 keep the positive charge off the carbon bearing $-$COOH, with C5 ultimately favored on steric grounds. That exercise was the moment EAS directing rules stopped being something I memorized and started being something I could derive from the resonance picture.

3. Key insights. The most useful realization was that the same nucleophilic acyl substitution I drew for base-catalyzed ester hydrolysis (Section C) is exactly what COX-1 carries out on aspirin in vivo, with Ser530's $-$OH playing the role of the nucleophile and salicylate playing the role of the leaving group. Once that link clicked, the pharmacology stopped being a separate list of clinical effects and collapsed into one Organic II mechanism applied to a much larger molecule. The corollary was that "drug design" and "reaction mechanism" are not separate disciplines for a molecule like aspirin – they are the same picture at different scales.

4. Personal growth. This was the first time I had to make every chapter of Organic I/II agree on a single molecule simultaneously, which is a very different exercise from solving a discrete homework problem. It pushed me to treat reactivity as one continuous story driven by structure rather than as independent textbook units, and to actually justify conclusions (e.g. EAS regiochemistry, leaving-group ability) instead of citing a rule of thumb. That habit of grounding every claim in a mechanism is what I will carry into future chemistry and biochemistry coursework.